Industrial Internet of Things (IIoT) sector. This AI revolution promises to reshape industries, but hardware development challenges persist, often following rigid, time-consuming paths such as the waterfall model, despite the availability of more agile methods.

With tight budgets and uncertainty in the R&D and proof-of-concept phases, hardware development can spiral into costly, ongoing loops due to evolving ideas and unclear requirements. Rapid technological evolution can render hardware obsolete upon commercialisation. The interplay between hardware and software results in debates on computation distribution, protocol placement, and collaboration between hardware and software teams. Unlike software, hardware changes late in development, incurring additional costs and delays. Technological advances outpace development, with processors and memory doubling every few years. Constant customer demand for the latest technology fuels the cycle of upgrades.

The SoM solution

Enter SoM, often referred to as a "computer on module." SoMs offer a modular approach to hardware development, akin to versatile Lego bricks that serve as foundational building blocks for various projects. Think of them as mini-computers equipped with processors, memory, Ethernet, Wi-Fi, and more, ready for customisation.

What sets SOMs apart is their generic nature; they are not application-specific. This versatility allows developers to create custom application boards tailored to specific industries or use cases while relying on a standardised SOM as the foundation. Whether you are working on an automotive project, an IoT application, or any other endeavor, SOMs offer modularity and flexibility.

One ongoing challenge in hardware development is resource allocation between hardware and software components. Deciding whether to handle computing at the edge or on the backend, choosing the placement of protocols, and resolving disputes between hardware and software teams are routine issues. SOMs have emerged as a game-changer to address these challenges and offer greater modularity and flexibility in hardware development. SoMs, or computers on modules (CoMs), provide a crucial bridge between hardware and software, offering numerous advantages that we will explore.

The anatomy of an SoM

An SoM is a comprehensive hardware solution comprising diverse components, such as processors, memory modules, connectivity interfaces like Ethernet and Wi-Fi, and audio codecs. Despite its inherent generic nature, an SoM serves as a foundational platform for crafting application-specific boards tailored to the unique requirements of a specific industry or use case. This strategy streamlines the hardware development process, notably reducing complexity and associated costs.

This prompts the question: Why opt for an SoM? There are several compelling reasons to choose an SoM as a core building block for your hardware development strategy. Here are some of the key advantages:

Modularity and flexibility. SoM separates core computing components from application-specific elements, allowing the creation of custom application boards while keeping the SoM consistent. This modularity fosters adaptability and future-proofing. While SoMs offer a generic starting point, they can be customised with additional peripherals or interfaces to meet the specific demands of your application.

Simplified development. SoM reduces the complexity of high-speed PCB design by providing pre-built modules. This streamlines development, reduces errors, and saves time, offering a simplified starting point for hardware development. It eliminates the need to design these elements from scratch.

Cost-efficiency. Custom boards often require more layers, which increases costs. In contrast, application boards using SoM can be simpler, more cost-effective, and quicker to develop. Leveraging an SoM takes advantage of economies

A strategic choice for embracing SoMs in hardware development

The decision to incorporate SoMs into hardware development projects is a strategic move driven by several critical factors. The key considerations that can simplify and enhance hardware development endeavors are:

Reduced supply chain complexity. Opting for SoMs streamlines the supply chain by benefiting from economies of scale. Manufacturers produce SoMs in bulk, ensuring consistent availability and simplifying procurement, reducing supply chain challenges, and enabling a focus on the project's core aspects.

Application-specific customisation. SoMs are a versatile foundation for tailoring application-specific boards, aligning hardware precisely with the needs. Custom boards enable unique features and superior performance, reducing development time and costs by leveraging SoMs as a sturdy base for specialised projects.

This contributes to a smoother and more cost-effective development process. This strategic choice streamlines development and positions success in delivering innovative and highly specialised hardware solutions.